

Problems and Exercises

- 2.0 Let P be a polyhedron and let F_0 be any nonempty face of P which is minimal with respect to inclusion. Taking $\mathbf{x}_0 \in F_0$, show that $F_0 - \mathbf{x}_0$ is a linear subspace, and that $F_0 - \mathbf{x}_0 = \text{lineal}(P)$.
Thus, if P has lineality space $\text{lineal}(P) = \{\mathbf{0}\}$, then every minimal nonempty face is a vertex — and in particular, P has a vertex.
- 2.1 Use the previous exercise combined with Theorem 1.2 to formulate and prove analogues of the Representation Theorem 2.15 for polyhedra. Special attention is needed for the formulation of part (5). In particular, what do you get in the case where P is a cone?
- 2.2 Show that every polytope is affinely isomorphic to a bounded intersection of an orthant with an affine subspace.
- 2.3 Construct a small poset that satisfies the conditions of Theorem 2.7 but does not correspond to a convex polytope. Does your example correspond to some geometric object?
- 2.4 Prove directly (i.e., without using polarity) that every face of a polytope P is contained in a facet.
- 2.5 If two 0/1-polytopes are combinatorially equivalent, does it follow that they are affinely isomorphic? (The answer is “no.”)
- 2.6 Let $f(d)$ be the number of combinatorial equivalence classes of d -dimensional 0/1-polytopes. The first values are $f(0) = f(1) = 1$, $f(2) = 2$, and $f(3) = 8$. Prove that $2^{2^{d-2}} < f(d) < 2^{2^d}$ for $d > 5$.
(This solves a problem of Billera & Sarangarajan [76, Sect. 3]. For the lower bound A. Sarangarajan and I suggest that you consider all the polytopes of the form $P = \text{conv}(S)$ for sets $S \subseteq \{0, 1\}^d$ that satisfy

$$\begin{aligned} \mathbf{x} \in S \text{ for all } \mathbf{x} \in \{0, 1\}^d \text{ with } x_d = 1, \\ \mathbf{0}, \mathbf{1} - \mathbf{e}_d \in S, \text{ and} \\ \mathbf{e}_1, \mathbf{1} - \mathbf{e}_d - \mathbf{e}_1 \notin S. \end{aligned}$$

There are $2^{2^{d-1}-4}$ such polytopes $P(S)$; show that their combinatorial equivalence classes are “small.”)

- 2.7 Assume that one is given the vertex-facet incidence matrix

$$M(P) \in \{0, 1\}^{m \times n} \equiv \{., *\}^{m \times n}$$

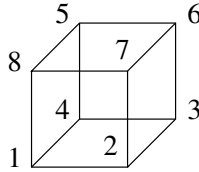
of a convex polytope P with m vertices and n facets.

How can the face lattice of the polytope P be uniquely reconstructed from the knowledge of $M(P)$ alone? How does the dimension of P appear in the computation? How does your algorithm fail if the matrix you apply it to is not the vertex-facet matrix of a polytope?

What is the relation between the matrices of P and P^Δ ?

2.8 Let P and P' be polytopes with vertex sets $V = \{v_1, \dots, v_n\}$ and $V' = \{v'_1, \dots, v'_n\}$. Assume that for every vertex set $F \subseteq V$ of a facet of P , the corresponding set $F' \subseteq V'$ is the vertex set of a facet of P' .

- (i) Show that for every face of P , the corresponding vertex set of P' forms a face in P' . Deduce that $\dim(P) \leq \dim(P')$.
- (ii) Lemma: if P and P' have the same dimension, then they are combinatorially equivalent under $v_i \rightarrow v'_i$.
(Hint: Assume this fails. Then P' has two facets, F'_1 and F'_2 , which are adjacent, such that of the corresponding vertex sets in P , the set F_1 forms a facet, but F_2 does not. (Here you can use the fact that the graph of $(P')^\Delta$ is connected: a proof is in the next lecture.) Now consider the ridge $F'_1 \cap F'_2$, which is a facet of F'_1 , and use induction on the dimension, applied to the polytopes F_1 and F'_1 .)
- (iii) Show that P and P' need not have the same dimension.
(Hint: For this, one can take a cube $P = C_3$, labeled as in the drawing, and a cyclic polytope $P' = C_4(8)$.)



- (iv) Assume that of a polytope P you are given the dimension, the vertex set, and a matrix $M(P) \in \{0, 1\}^{m \times n}$ such that every row of $M(P)$ represents a facet of P .
How can you tell whether this list of facets is complete?
(Remark: this is not too easy; one can use tools from Chapter 8, or from homology theory.)
- (Part (ii) is important: See Klee & Minty [330, p. 167], Amenta & Ziegler [17], and elsewhere. (iii) points to an error in [330, p. 167].)
- 2.9 Define the *face figure* P/F for any face of P by $P/F := (F^\diamond)^\Delta$, that is, a polar of the face of P^Δ which corresponds to F . (The face figures P/F are also known as the *quotients* of P . Thus a quotient of P is the same thing as an iterated vertex figure.)

Show that this is a polytope of dimension

$$\dim(P/F) = \dim(P) - \dim(F) - 1.$$

Characterize the face lattice of a face figure in terms of the face lattice of P and of the element $F \in L(P)$.

Describe a more direct construction of P/F , generalizing the case of a vertex figure. How can the face figure P/F be obtained as an iterated vertex figure?

- 2.10 In conditions (iii) and (iv) of the characterization of the interior of a polytope (Lemma 2.8), can we assume the $\mathbf{x}_i$ to be vertices?
- 2.11 Show that if $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subseteq \text{vert}(P)$ is a set of vertices of P , then $F = \{\mathbf{v}_1\} \vee \dots \vee \{\mathbf{v}_k\}$ holds in $L(P)$ if and only if $\frac{1}{k} \sum_{i=1}^k \mathbf{v}_i \in \text{relint}(F)$. Generalize to get a formula for the join of a set of faces $\{G_1, \dots, G_k\} \subseteq L(P)$.
- 2.12 Compute directly that every polar of a simplex is a simplex.
- 2.13 Define the polar of a cone by

$$C^\Delta := \{\mathbf{c} \in (\mathbb{R}^d)^* : \mathbf{c}\mathbf{x} \leq 0 \text{ for all } \mathbf{x} \in C\}.$$

Show that this definition (with “0” instead of “1”) is a special case of our definition for arbitrary subsets.

Formulate and prove the analogs of our Theorems 2.11 and 2.12.

- 2.14 Let P be a d -polytope in $\mathbb{R}^d$, given by the system

$$P = P(A, \mathbf{z}).$$

An inequality in this system is called *redundant* if deleting it from the inequality system $X\mathbf{x} \leq \mathbf{z}$ does not change the polyhedron; otherwise the inequality is called *irredundant*.

- (i) Derive from Farkas lemma III that an inequality is redundant if and only if it can be written as a positive combination of other inequalities in the system.
- (ii) Derive from a Farkas lemma that if $P = P(A, \mathbf{z}) \neq \emptyset$ and if none of the inequalities of a system $A\mathbf{x} \leq \mathbf{z}$ is redundant, then each of them defines a facet. Thus, every polytope is the intersection of its facet-defining inequalities.
- (iii) Show that if $\mathbf{x}_F \in \text{relint}(F)$ is a point in the relative interior of a facet $F \in L(P)$, then the inequality $\mathbf{a}\mathbf{x} \leq z$ defines the facet F if and only if it is valid for P , and $\mathbf{a}\mathbf{x}_F = z$.
- (iv) Show that for every facet F of P , there is a unique inequality $\mathbf{a}\mathbf{x} \leq z$ which defines F , and for which $\sum_{i=1}^d |a_i| = 1$.

Together, these statements prove that any irredundant description of a d -polytope $P \subseteq \mathbb{R}^d$ as an $\mathcal{H}$ -polytope contains exactly one inequality for each facet of P .

Show that this statement could also be derived by polarization of Proposition 2.2.

What happens in the situation when $\dim(P) < d$? How much of the uniqueness statement in part (iii) can you rescue?

- 2.15 Let $P = \text{conv}(V) \subseteq \mathbb{R}^d$ be a convex d -polytope, and assume that some description of P as an $\mathcal{H}$ -polytope is known,

$$P = P(A, \mathbf{z}).$$

With every inequality $\mathbf{a}_i \mathbf{x} \leq z_i$ in this system, associate its vertex set

$$V_i := \{\mathbf{v} \in V : \mathbf{a}_i \mathbf{v} = z_i\}.$$

Show that the following criteria can be used to check whether an inequality is redundant.

- (i) The inequality $\mathbf{a}_i \mathbf{x} \leq z_i$ is redundant if and only if $V_i \subseteq V_j$ for some $j \neq i$.
- (ii) If $V_i = V_j$, then either the inequalities are multiples of each other, or they can both be deleted from the system.
- (iii) An inequality is irredundant if and only if it defines a facet of P and no multiple of it is contained in the system.
- (iv) If $|V_i| < d$, then the inequality $\mathbf{a}_i \mathbf{x} \leq z_i$ is redundant.
- (v) The inequality $\mathbf{a}_i \mathbf{x} \leq z_i$ is irredundant if and only if there is no multiple of it in the system, and the rank of the matrix given by $\left\{ \begin{pmatrix} 1 \\ \mathbf{v} \end{pmatrix} : \mathbf{v} \in V_i \right\}$ is d .

(Parts (i) and (v) yield complete criteria for redundancy, which can be checked explicitly. Note that there was no assumption that the set V has to be minimal. Condition (i) is (equivalent to) the main condition of Chernikova [149], while condition (v) is a rank test that seems not so efficient. For example, a combination of criteria (i) and (iv) makes sense in practice.)

- 2.16 Let $P = \text{conv}(V) \subseteq \mathbb{R}^d$ be a convex d -polytope, and assume that an irredundant description of P as an $\mathcal{H}$ -polytope,

$$P = P(A, \mathbf{z}),$$

is known, such that the inequalities $\mathbf{a}_i \mathbf{x} \leq z_i$ describe all the distinct facets of P , without duplication. For every inequality $\mathbf{a}_i \mathbf{x} \leq z_i$ in the system, let V_i be the set of points in V which satisfy it with equality. Show that from this, an irredundant description of $\text{proj}_d(P) \subseteq \mathbb{R}^{d-1}$ can be obtained from the following criteria:

- (i) if $a_{id} = 0$, then $\mathbf{a}_i \mathbf{x} \leq z_i$ determines a facet of $\text{proj}_d(P)$,
- (ii) if $a_{id} > 0$ and $a_{jd} < 0$, then the inequality

$$a_{id}\mathbf{a}_j + (-a_{jk})\mathbf{a}_i \leq a_{id}z_j + (-a_{jk})z_i$$

defines a facet of $\text{proj}_d(P)$ if and only if $V_i \cap V_j \subseteq V_k$ holds for no $k \neq i, j$.

Furthermore, show that if $|V_i \cap V_j| < d - 1$, then the combined inequality in (ii) is redundant. In particular, this is the case if $d \geq 2$ and $V_i \cap V_j = \emptyset$. (Give geometric proofs — they are easier than algebraic ones!)

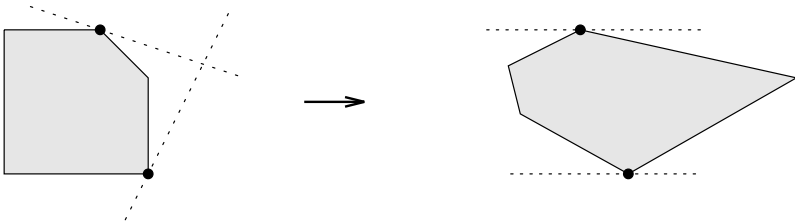
Explain how, by using Fourier-Motzkin elimination (Theorem 1.4) together with these redundancy criteria, one can obtain a complete irredundant description of $P := \text{conv}(V) \subseteq \mathbb{R}^d$, even if the set V contains more points than just the vertices of P .

How can the criterion be adapted for the case of polyhedra, where the input is a polyhedron given as $P = \text{conv}(V) + \text{cone}(X)$?

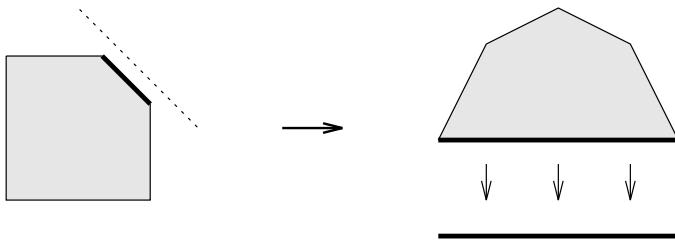
What happens in the situation where $\dim(P) < d$, and how can the difficulty there be overcome?

(The necessary and sufficient criterion can be found, for example, in Burger's version [140, Thm. 3] of the double description method [416]. The test on " $V_i \cap V_j = \emptyset$ " is a heuristic in Chernikova [149].)

- 2.17 Show that if $P \subseteq \mathbb{R}^d$ is a polytope with two distinct vertices $\mathbf{u}, \mathbf{v}$, then there is a projective transformation $P \rightarrow P'$ such that the vertices $\mathbf{u}'$ and $\mathbf{v}'$ have the smallest, respectively the largest, x_d -coordinate among all vertices of P' .



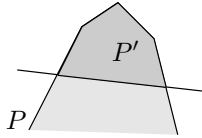
- 2.18 Let $P \subseteq \mathbb{R}^d$ be a polytope, let F be a facet of P , and let $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^{d-1}$ be a projection map (for example, deleting the last coordinate). By “moving a point beyond F to infinity,” show that by a projective transformation $P \rightarrow P'$, we can obtain $\pi(P') = \pi(F')$ and $\pi^{-1}(G') \cap P' = \pi^{-1}(G') \cap F'$ for every proper face $G' \subset \pi(F')$.



- 2.19 Using a Farkas lemma, show that for every unbounded pointed polyhedron P there is an inequality $\mathbf{a}\mathbf{x} \leq 1$ such that

$$P' := \{\mathbf{x} \in P : \mathbf{a}\mathbf{x} \leq 1\}$$

is a polytope with a facet $F' := \{\mathbf{x} \in P : \mathbf{a}\mathbf{x} = 1\}$, such that the k -faces of F' correspond to the unbounded $(k+1)$ -faces of P , and the k -faces of P' that are not faces of F' are in bijection with the k -faces of P .



Show that a polyhedron combinatorially equivalent to P' can also be obtained from P by taking the closure of the image of P after a “nonadmissible” projective transformation that moves the face at infinity into $\mathbb{R}^d$.

How can this be used to study the combinatorics of pointed unbounded polyhedra in terms of “polytopes with a distinguished face”?

- 2.20 Let $P \in \mathbb{R}^d$ be a d -polyhedron with n facets and at least two vertices, and assume that $\mathbf{0} \in \text{int}(P)$. With P , associate the new polyhedron

$$P^\circ := (\text{conv}(\text{vert}(P^\Delta) \setminus \{\mathbf{0}\}))^\Delta.$$

Show that if P is a polytope, then $\mathbf{0}$ is an interior point of P^Δ , and $P^\circ = P$.

If P is unbounded, then $\mathbf{0}$ is on the boundary of P^Δ . Show that the polar, with respect to an interior point of $\text{conv}(\text{vert}(P^\Delta) \setminus \{\mathbf{0}\})$, is a d -polytope P° with n facets and with more vertices than P .

- 2.21 The *Carathéodory curve* [142] in $\mathbb{R}^{2d}$ is given by

$$\mathbf{y} : \mathbb{R} \longrightarrow \mathbb{R}^{2d}, \quad u \longmapsto \mathbf{y}(u) := \begin{pmatrix} \cos(u) \\ \sin(u) \\ \cos(2u) \\ \sin(2u) \\ \vdots \\ \cos(du) \\ \sin(du) \end{pmatrix}.$$

- (i) Show that for $0 \leq u_1 < u_2 < \dots < u_n < 2\pi$ with $n > 2d$, the convex hull

$$C'_{2d}(u_1, u_2, \dots, u_n) := \text{conv}\{\mathbf{y}(u_1), \mathbf{y}(u_2), \dots, \mathbf{y}(u_d)\}$$

is combinatorially equivalent to the cyclic polytope $C_{2d}(n)$.

(Hint: You will find a useful hint in Grünbaum [252, p. 67, Ex. 23].)

(ii) Prove that in the case $d = 2$ the map:

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} \longmapsto \frac{1}{3 + 4y_1 + y_3} \begin{pmatrix} 2y_2 + y_4 \\ 1 - y_3 \\ 2y_2 - y_4 \\ 3 - 4y_1 + y_3 \end{pmatrix}$$

is a projective transformation, which maps (the relevant part of) the Carathéodory curve to the moment curve.

Conclude that for suitable real parameters $t_1 < t_2 < \dots < t_n$, the polytope $C'_4(u_1, \dots, u_n)$ is in fact *projectively equivalent* to the “standard” cyclic polytope $C_4(t_1, \dots, t_n)$ defined via the moment curve (Example 0.6).

(In fact, for general $d \geq 1$ the polytope $C'_{2d}(u_1, u_2, \dots, u_n)$ is projectively equivalent to some standard cyclic polytope $C_{2d}(t_1, \dots, t_n)$. For this one can explicitly construct a projective transformation that takes the Carathéodory curve to the moment curve, using a substitution of the type

$$t := \frac{1 - \cos(u)}{\sin(u)} = \frac{\sin(u)}{1 + \cos(u)}$$

and some elementary trigonometric identities, such as the formulas $\sin(2t) = 2 \sin(t) \cos(t)$ and $\cos(2t) = 2 \cos(t)^2 - 1$.)

2.22 For relatively prime natural numbers $p, q \in \mathbb{N}$ (i.e., numbers with no common factor), define the *bicyclic polytope* $P_4(p, q, n)$ as the convex hull of the $n \geq 5$ points

$$\mathbf{v}_i := \begin{pmatrix} \cos(2p\pi \frac{i}{n}) \\ \sin(2p\pi \frac{i}{n}) \\ \cos(2q\pi \frac{i}{n}) \\ \sin(2q\pi \frac{i}{n}) \end{pmatrix}$$

for $1 \leq i \leq n$.

Describe the geometry and combinatorics of these polytopes.

- (i) What symmetries do they have? Show that all the facets of $P_4(p, q, n)$ are combinatorially equivalent.
- (ii) How many facets does $P_4(p, q, n)$ have? Find the conditions on p , q , and n under which $P_4(p, q, n)$ is simplicial.
- (iii) Using part (i) or (ii) of the previous exercise, show that the 4-polytopes $P(1, 2, n)$ and $P(1, \frac{n-1}{2}, n)$ (for odd n) are combinatorially equivalent to cyclic polytopes $C_4(n)$.

(Smilansky [502, 503])

2.23 Try to estimate the number of combinatorial equivalence classes of d -dimensional polytopes with n vertices.

(Goodman & Pollack [235, 236], Alon [12])

3

Graphs of Polytopes

The vertices and edges of a d -polytope P form an undirected graph $G(P)$ that encodes a lot, but not everything, about the combinatorial structure of the polytope.

In this lecture we discuss three fundamental topics about graphs of polytopes: the monotone Hirsch conjecture (for which we prove validity for $0/1$ -polytopes, and Kalai's recent bound for the general case), Kalai's reconstruction of simple polytopes, and Balinski's d -connectivity theorem.

Before we look into this, we will establish two technical tools: the power and the glory of half-sentences like “let L be a line in general position,” and the (geometric version of the) simplex algorithm for linear programming, which is the most fundamental search technique on polytopes.

3.1 Lines and Linear Functions in General Position

We start with a short discussion of the concept of “general position with respect to P ” — this will yield a few useful tools for proofs in subsequent sections. To illustrate this, we will sketch a quick-and-dirty geometric version of linear programming in the next section.

With all the work we did in Lectures 1 and 2, we can now assume that we are dealing with a d -dimensional polytope P in $\mathbb{R}^d$, in which $\mathbf{0}$ is an interior point. We alternate freely between representations in terms of vertices, $P = \text{conv}(V)$, and in terms of facets, $P = P(A, \mathbf{1})$. Here $A \in \mathbb{R}^{n \times d}$ is a matrix that is considered as a set of rows, $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$.